



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Graph for Navigation and Shortest Path

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Course: Data Structure

Introduction

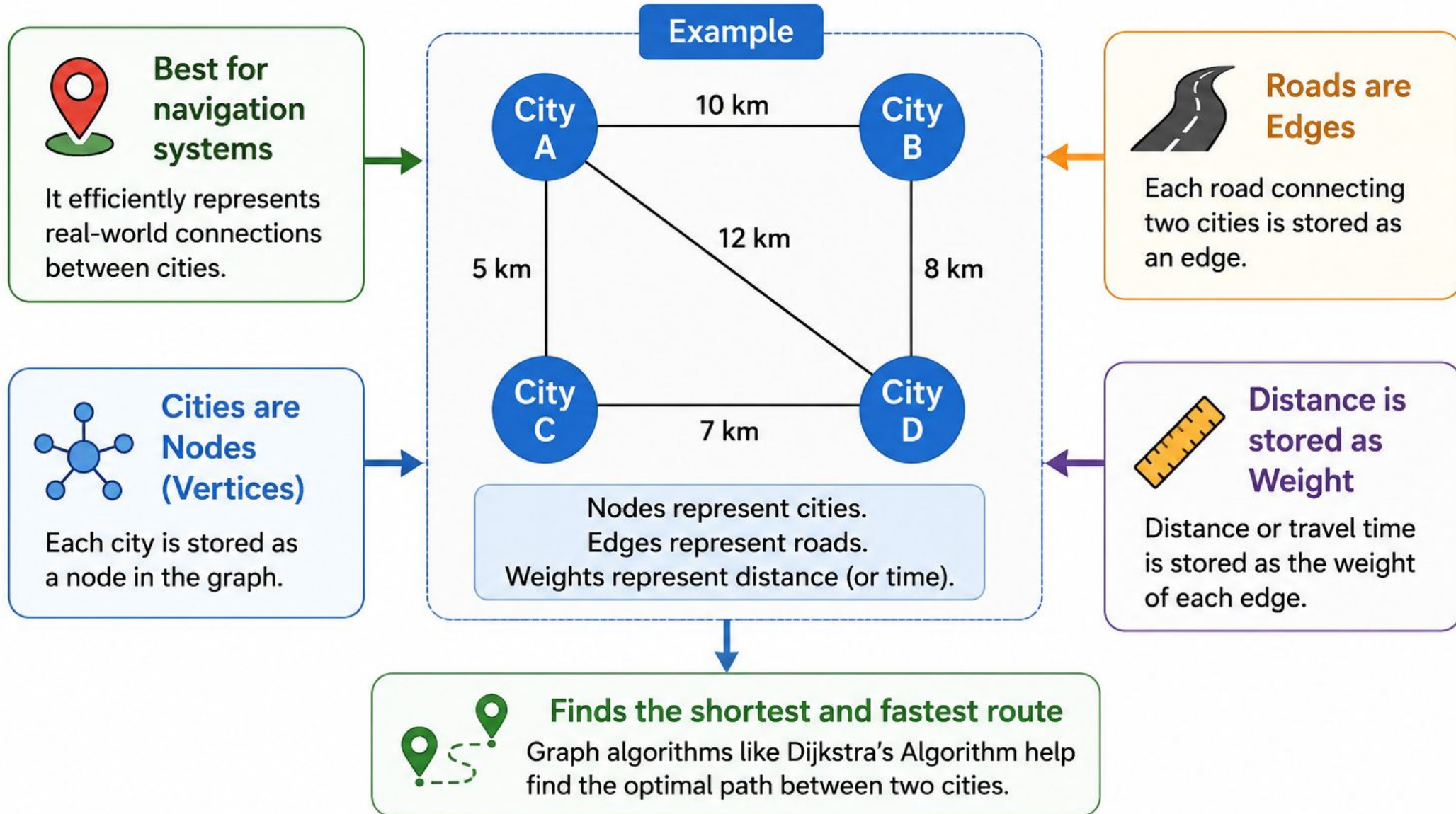
- A navigation system helps us travel from one place to another.
- It shows different roads connecting cities or locations.
- The system finds the shortest and fastest route.
- A Graph data structure is used to store these connections.
- This makes navigation faster, easier, and more accurate.

Problem Statement

- A navigation system must store many cities and roads.
- It should find the shortest route between two locations.
- It should provide the fastest and most efficient path.
- It must handle multiple routes between cities.
- Choosing the right data structure is important for quick and accurate navigation.

Suitable Data Structure – Graph

A Graph is the best data structure for a navigation system.



Graph Representation



Vertices (Nodes) represent cities.



Edges represent the roads connecting cities.



Weights show the distance or travel time between cities.



The graph stores all city connections clearly.



It helps find the shortest route efficiently.



→ Vertices (Nodes) = Cities

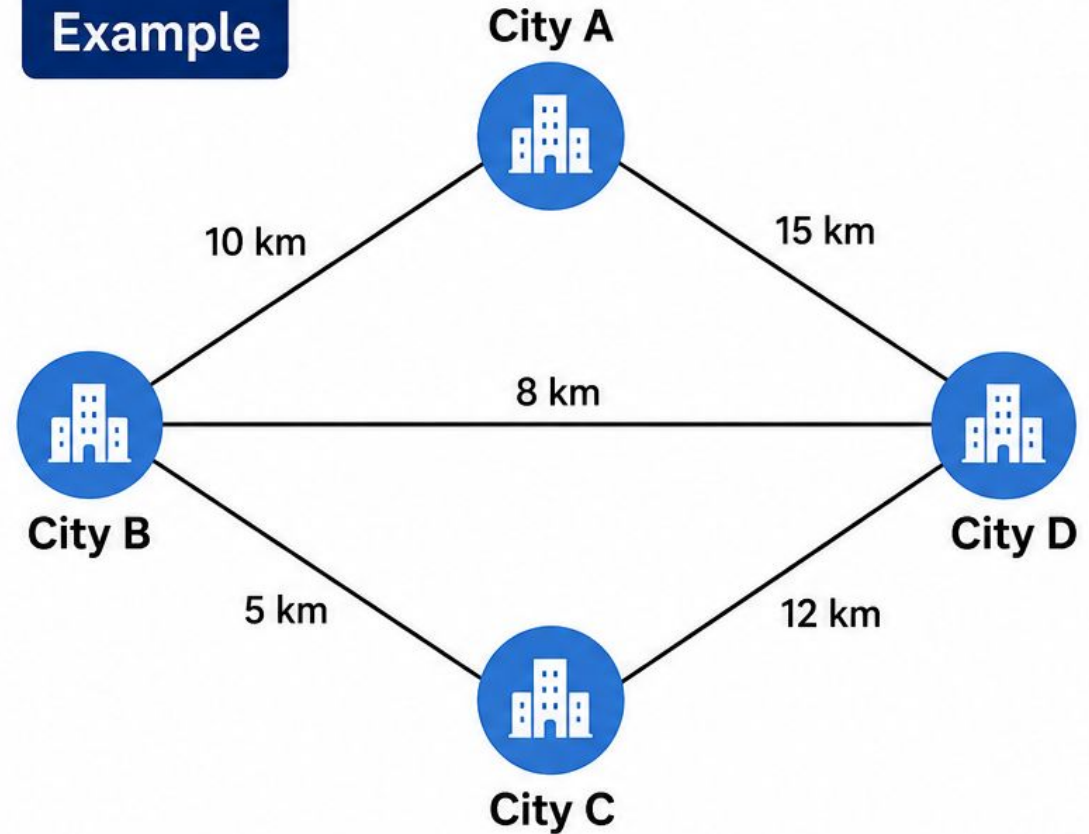


→ Edges = Roads

10 km

→ Weights = Distance / Travel Time

Example



This graph helps the navigation system find the shortest and most efficient route between any two cities.

Conclusion

- A Graph is the best data structure for a navigation system.
- It represents cities as nodes and roads as edges.
- It stores distance or travel time as weights.
- It helps find the shortest and fastest route efficiently.
- Therefore, graphs make navigation systems smart, fast, and reliable.